Spannovation

Aaryan Sehgal B.Eng.

Junior Bridge Designer

Aaryan is a Civil Engineering graduate from Chandigarh College of Engineering and Technology, affiliated with Punjab University. He is a hardworking individual with a keen interest in designing bridge infrastructure projects. As a fresher with one year of experience, he has gained a remarkable understanding of complex bridge structures and geometry. He has also developed a strong interest in Visual Basic for Applications (VBA) and has developed Excel macro programs to automate tasks, reduce design time, and enhance the quality of consultancy services. His analytical skills and ability to collaborate effectively within a team make him a valuable asset to any engineering project.

Expertise

Construction Engineering
Steel Structures
Programming

Past Firms

Mortar Civil Solutions Private Limited,
Junior Bridge Designer, 2023 - 2024

Software Skills

MIDAS Civil, STAAD Pro, IdeaStatiCa, AutoCAD, Struct Assist, CAD Assist, SAP2000, Microsoft Excel

Education

Chandigarh College of Engineering and Technology, Panjab University
Bachelor of Engineering, 2023

Erection Engineering

New Pattullo Bridge Replacement Project, Vancouver, Canada
Junior Bridge Designer | 2023 - Present | $1.3B | B.C. Ministry of Transportation
He participated in the construction of the New Pattullo Stay Cable Bridge with a main span of 332m. He conducted detailed erection analysis of end span girder segments from the S2 to S3 piers and designed rigging arrangements for these segments, including slings and lifting mechanisms. Additionally, he designed temporary supports for the segments on the S2 and S3 piers. He created and analyzed finite element models of steel box girder segments for counterweight casting of concrete pours and pre-cast concrete panels. He was involved in the detailed design of various components, such as the Access Platform for the S2 pier cap, end span superstructure temporary bracings, Floor Beam Temporary Strut, temporary shoring tower (TP3), and the assembly for lowering formwork of the S2 and S3 pier caps, which were designed by DOKA. He also assisted with minor tasks related to Pier Table erection. Moreover, he developed spreadsheets to check the capacity of both temporary and permanent structural members according to CSA S6:19 code and verified construction drawings for accuracy and compliance.

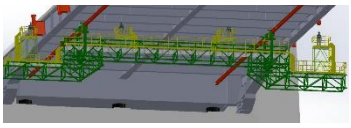
New Athabasca River Bridge Project, Alberta, Canada
Junior Bridge Designer | 2023 - Present | ~$73M | Alberta Transportation
He participated in the construction of the three-span (80m + 99m + 80m) bridge with four girder lanes. He is involved in the erection of steel I-Girders and conducted detailed construction stage erection analysis of the girder segments, designing rigging arrangements including slings and lifting mechanisms. Additionally, he designed temporary restraints for the segments on the piers, abutments, and temporary towers to restrict the lateral movement of the girders.

Queen Elizabeth II (QEII) & 65th Avenue Interchange Project, Leduc, Canada
Junior Bridge Designer | 2023 - 2024 | ~$93M | Government of Alberta
He participated in the construction of the three-span (51m + 51m + 65m) overpass bridge with six girder lanes for the project. He was involved in the erection of steel I-Girders, facing the main challenge of height restrictions due to the nearby airport. He conducted detailed construction stage erection analysis of the girder segments and designed rigging arrangements, including slings and lifting mechanisms. Additionally, he designed temporary restraints for the segments on the piers, abutments, and temporary towers to restrict the lateral movement of the girders.

Strathcona County Footbridge Project, Edmonton, Canada
Junior Bridge Designer | 2023 - 2024 | ~$42M | Government of Alberta
He participated in the construction of the three-span steel girder segments with a total bridge length of approximately 230 meters. He was involved in the erection of steel tub girders, facing the main challenge of the unique S-shaped curved geometry of the bridge. He conducted detailed construction stage erection analysis of the girder segments and designed rigging arrangements, including slings and lifting mechanisms. Additionally, he designed temporary restraints for the segments on the piers, abutments, and temporary towers to restrict the lateral and torsional movement of the girders.

Wabasca River Bridge Project, Alberta, Canada
Junior Bridge Designer | 2023 | ~$57M | Alberta Transportation
He participated in the construction of a two-span bridge with four girder lanes of 77 meters each. He was involved in the erection of steel I-Girders and conducted detailed design of rigging arrangements, including slings, and lifting mechanisms. Additionally, he designed temporary restraints for the segments on the piers, abutments, and temporary towers to restrict the lateral movement of the girders.

Steel Structures

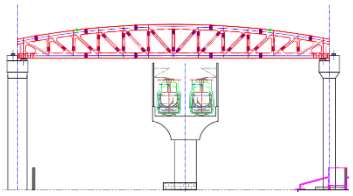


Pattullo Bridge Permanent Gantry Project, Vancouver, Canada
Junior Bridge Designer | 2023 - 2024 | B.C. Ministry of Transportation
He participated in the evaluation of the structural integrity of steel truss gantry elements and their applicable connections. This self-propelled gantry spanning two meters in width and 40 meters in length is utilized under the new Pattullo Bridge to facilitate inspection, maintenance, repairs, and repainting works. He was involved in modeling and analyzing various joint connections within the truss gantry

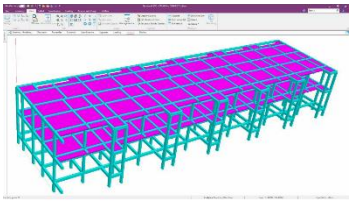
using the FEA software IdeaStatiCa and checked various pin connections according to the CSA S6:19 code.

Other Projects

Greater Male Connectivity- Malé to Thilafushi Link Project, Maldives
Junior Bridge Designer | 2023 | $500M | Government of Maldives
He participated in quantity estimation for both the superstructure and substructure of Module 17, which spans 337 meters with a span arrangement of 96m - 148m - 93m. He was involved in meticulous calculations of bar bending schedules and consideration of all relevant factors to produce accurate estimations



Madhuban Chowk Steel Truss Bridge Project, Delhi, India
Intern | 2023 | Delhi Metro Rail Corporation Limited
He participated in the construction of a 50m span steel truss bridge. He was involved in analyzing the trestle supports for the bridge segments by modelling the structure in MIDAS Civil software.



Duplex Designing based on Sustainable Low-Cost Methodology
Student | 2022 | Chandigarh College of Engineering and Technology
He participated in a group project with his classmates during the seventh semester of his bachelor's course. In this project, they analyzed the structure of six duplexes both manually and using STAAD.Pro software. Additionally, they estimated and analyzed the project's cost by employing sustainable and low-cost materials.